

Classical Computer Security Models

A Compact Research Guide for Cybersecurity Students

Introduction

Classical security models form the theoretical foundation of modern cybersecurity. Although developed between the 1970s and 1990s, they continue to influence contemporary technologies such as Identity and Access Management (IAM), Role-Based Access Control (RBAC), Attribute-Based Access Control (ABAC), Zero Trust Architecture, database security, cloud security, and secure operating systems.

While security analysts rarely implement these models directly, understanding them provides insight into **why modern access control systems work as they do**. Many cybersecurity certifications—including CompTIA Security+, ISC2 Certified in Cybersecurity (CC), CISSP, GIAC, and university programs—cover these models because they introduce the principles behind confidentiality, integrity, conflict-of-interest prevention, privilege management, and secure information flow.

1. Brewer–Nash Model (Chinese Wall Model)

Authors: David F. C. Brewer and Michael J. Nash (1989)

Purpose

The Brewer–Nash Model prevents **conflicts of interest** by dynamically restricting access to sensitive information based on a user's previous data access.

Core Idea

Once a user accesses confidential information belonging to one company, they are automatically prevented from accessing confidential information belonging to competing companies within the same conflict-of-interest class.

Example

A cybersecurity consultant working with **Company A** cannot later access confidential documents belonging to **Company B**, a direct competitor.

Strengths

- Prevents industrial espionage.
- Supports ethical information handling.

- Dynamatically adapts to user activity.

Weaknesses

- Complex to administer in large organizations.
- Requires careful classification of competing entities.

Modern Applications

- Consulting firms
- Investment banks
- Law firms
- Audit companies
- Managed Security Service Providers (MSSPs)

Why Beginners Should Learn It

Modern Identity and Access Management systems often implement similar conflict-of-interest controls. Understanding this model helps analysts appreciate how organizations enforce ethical access policies.

2. Goguen–Meseguer Model

Authors: Joseph A. Goguen and José Meseguer (1982)

Purpose

The model introduces the concept of **noninterference**, ensuring that actions performed at higher security levels cannot influence what lower-level users observe.

Core Idea

Information must never leak indirectly between different security domains.

Example

A classified military system should not reveal secret information through timing differences, resource usage, or system behavior observable by lower-privileged users.

Strengths

- Strong theoretical confidentiality guarantees.
- Protects against covert information leakage.
- Influential in secure operating system design.

Weaknesses

- Difficult to implement perfectly.

- Mainly theoretical.

Modern Applications

- Military systems
- Government systems
- Secure kernels
- High-assurance operating systems

Why Beginners Should Learn It

Modern discussions about side-channel attacks, covert channels, and information leakage stem directly from ideas introduced by this model.

3. Sutherland Model

Author: David Sutherland (1986)

Purpose

The Sutherland Model protects **information integrity** by preventing improper information flow that could corrupt trusted data.

Core Idea

It controls how information moves between subjects and objects to maintain trustworthy data.

Example

An untrusted process should never overwrite or influence verified financial records.

Strengths

- Focuses on preserving integrity.
- Supports trusted computing.

Weaknesses

- Less widely adopted than other integrity models.
- Limited practical implementations.

Modern Applications

- Critical infrastructure
- Financial systems
- Safety-critical environments

Why Beginners Should Learn It

Integrity is one of the three pillars of the CIA Triad. This model illustrates how protecting data from unauthorized modification is just as important as preventing unauthorized disclosure.

4. Graham–Denning Model

Authors: Graham and Peter J. Denning

Purpose

The Graham–Denning Model formalizes **secure management of subjects, objects, and permissions**.

Core Idea

It defines eight secure operations for creating, deleting, assigning, and transferring ownership and access rights.

These operations include:

- Create subject
- Delete subject
- Create object
- Delete object
- Grant rights
- Remove rights
- Transfer ownership
- Change permissions securely

Strengths

- Strong administrative access control.
- Clearly defines privilege management.

Weaknesses

- Limited focus on dynamic environments.
- Does not address modern distributed systems.

Modern Applications

- Identity Management
- Active Directory administration
- Cloud permission management

- Enterprise IAM

Why Beginners Should Learn It

Many administrative operations performed by system administrators today—creating users, assigning permissions, delegating ownership—follow principles established by this model.

5. Harrison–Ruzzo–Ullman (HRU) Model

Authors: Michael A. Harrison, Walter L. Ruzzo and Jeffrey D. Ullman (1976)

Purpose

The HRU Model studies how permissions evolve over time and whether a protection system can remain secure as rights change.

Core Idea

It introduced the **Safety Problem**:

Can a system guarantee that unauthorized access rights will never eventually appear?

The researchers proved that determining this is **undecidable** in the general case.

Strengths

- Foundation of access control theory.
- Influenced modern authorization systems.
- Important in formal verification research.

Weaknesses

- Highly theoretical.
- Difficult to apply directly in production environments.

Modern Applications

- Formal verification
- Operating system security
- Access control research
- Authorization engines

Why Beginners Should Learn It

The HRU Model demonstrates why access control becomes increasingly difficult as systems grow. Understanding permission propagation helps analysts avoid privilege escalation and excessive permissions in enterprise environments.

Comparison

Model	Primary Goal	Modern Equivalent
Brewer–Nash	Prevent conflicts of interest	Ethical access control, consulting, finance
Goguen–Meseguer	Prevent information leakage	Secure kernels, Zero Trust concepts, MLS
Sutherland	Preserve integrity	Trusted computing, critical infrastructure
Graham–Denning	Secure administration	IAM, RBAC, Active Directory
Harrison–Ruzzo–Ullman	Permission safety	Authorization systems, formal verification

Why These Models Matter Today

Although modern cybersecurity relies on technologies such as Zero Trust, cloud-native security, endpoint detection and response (EDR), and identity-centric architectures, these classical models continue to underpin many of their principles. Concepts like least privilege, separation of duties, secure permission management, prevention of information leakage, and integrity protection can all be traced back to this foundational research.

For aspiring cybersecurity professionals, these models help explain **why** access control systems are designed the way they are—not just **how** to configure them. This theoretical understanding is valuable for roles including SOC Analyst, Security Engineer, IAM Engineer, GRC Analyst, Security Architect, OT/ICS Security Specialist, and Penetration Tester, and it is frequently assessed in certifications such as CompTIA Security+, ISC2 CC, CISSP, and GIAC.

Research Methodology

This report was prepared using a structured literature review methodology. Information was synthesized from peer-reviewed academic publications, foundational computer security literature, and authoritative guidance published by organizations such as the National Institute of Standards and Technology, Cybersecurity and Infrastructure Security Agency,

European Union Agency for Cybersecurity, and the International Information System Security Certification Consortium. Classical security models were analyzed in their historical context and then related to contemporary cybersecurity practices, including Identity and Access Management (IAM), Role-Based Access Control (RBAC), Attribute-Based Access Control (ABAC), Zero Trust Architecture, and secure operating system design. The objective was to produce an accurate, vendor-neutral summary that is suitable for educational use and aligns with current industry best practices.